

At its core, **Document Overload Automation** addresses a simple, massive real-world problem: **organizations are drowning in paperwork, and processing it manually takes way too much time and effort.**

Here is the breakdown of what this SIH problem statement means, why it exists, and what you are expected to build.

1. The Real-World Problem (Why this exists)

Imagine a government office, a bank, or a large university receiving **thousands of documents every day**—PDFs, scanned physical forms, hand-written notes, emails, invoices, and IDs.

Currently, human employees have to:

- Open every single file manually.
- Read it to figure out what type of document it is (e.g., is it an application, a complaint, or a tax form?).
- Type key details (names, dates, reference numbers, amounts) into a system manually.
- Route it to the right department or flag errors.

This leads to **massive backlogs, high operational costs, human errors, and frustratingly slow turnaround times.**

2. What the Problem Statement Asks You to Do

Your goal is to build an **Intelligent Document Processing (IDP) System** using AI/ML that automates this entire lifecycle from start to finish.

In plain English, your software should be able to:

- 📁 **Intake & Read:** Accept documents in any format (scanned images, multi-page PDFs, low-quality photos) and read the text using **Optical Character Recognition (OCR)**.
- 🏷️ **Classify:** Automatically identify the type of document (e.g., "This is a PAN card," "This is a medical bill").
- 🔍 **Extract Key Data:** Pull out important fields without human help (e.g., extract Name, Date of Birth, Total Amount, Account Number).
- 📝 **Summarize & Verify:** Use Large Language Models (LLMs) or NLP to summarize long documents, check for missing information, or flag suspicious/fraudulent entries.
- ⚙️ **Route & Integrate:** Automatically push the structured data into a database or trigger the next step in a business workflow.

3. Key Components You'll Need to Build

To win or excel in this challenge, your solution typically needs to cover these 4 building blocks:

Phase	What Your System Does	Typical Tech / Tools
1. Digestion (OCR)	Converts image/scanned files into readable digital text.	Tesseract, AWS Textract, PaddleOCR, EasyOCR
2. Understanding	Contextual understanding,	SpaCy, HuggingFace

(NLP/LLM)	classification, and field extraction.	Transformers, LangChain, OpenAI API / Llama
3. Validation & Rules	Checks if extracted data is valid (e.g., valid date format, non-empty fields).	Python rules, RegEx, Pydantic
4. Dashboard & Workflow	A clean UI for humans to review flagged low-confidence documents.	React/Vue frontend, Streamlit, FastAPI/Django backend

4. The "Winning Edge" for SIH

Judges for document automation problems are usually looking for solutions that handle **edge cases**:

1. **Multilingual / Indic Languages:** Handling Hindi or local regional scripts alongside English.
2. **Handwritten or Poor Quality Documents:** Cleaning up noisy images before OCR.
3. **Human-in-the-Loop (HITL):** A simple review dashboard where a human can verify confidence scores below 85% with a single click.
4. **Data Privacy & Security:** On-premise or privacy-focused AI deployment (using open-source models like Llama-3 or Mistral).